Experiment 3 - Lists and Tuples

Python Lab Manual — Program and Output

EX. NO: 3(a) — Print items present in a library

PROGRAM

```
library = ["books", "author", "barcodenumber", "price"]
library[0] = "Let us C"
print(library[0])
library[1] = "Yashavant Kanetkar"
library[2] = 123987
library[3] = 234
print(library)

tup1 = (12134, 250000)
tup2 = ("books", "totalprice")
tup3 = tup1 + tup2
print(tup3)
```

OUTPUT

```
Let us C
['Let us C', 'Yashavant Kanetkar', 123987, 234]
(12134, 250000, 'books', 'totalprice')
```

EX. NO: 3(b) — Print system names

PROGRAM

```
print("\nLISTS:")
Computers = ["Dell", "Lenovo", "HP", "Apple", "Acer", "Asus"]
new_list = []

for i in Computers:
    if "A" in i:
        new_list.append(i)
        print(new_list)

print("\nTUPLE:")
Computers = ("Dell", "Lenovo", "HP", "Apple")
print(Computers)
for computer in Computers:
    print(computer)
```

OUTPUT

```
LISTS:
['Apple']
['Apple', 'Acer']
['Apple', 'Acer', 'Asus']

TUPLE:
('Dell', 'Lenovo', 'HP', 'Apple')
Dell
Lenovo
HP
Apple
```

EX. NO: 3(c) — Materials required for construction of a building

PROGRAM

```
print("\nLISTS:")
materials = ["cementbags", "bricks", "sand", "Steelbars", "Paint"]
materials.append("Tiles")
materials.insert(3, "Aggregates")
materials.remove("sand")
materials[5] = "electrical"
print(materials)

print("\nTUPLES:")
```

```
materials = ("cementbags", "bricks", "sand", "Steelbars", "Paint")
print(materials)
print("list of element is=", materials)
print("materials[0]:", materials[0])
print("materials[1:3]:", materials[1:3])
```

OUTPUT

```
LISTS:
['cementbags', 'bricks', 'Aggregates', 'Steelbars', 'Paint', 'electrical']

TUPLES:
('cementbags', 'bricks', 'sand', 'Steelbars', 'Paint')
list of element is= ('cementbags', 'bricks', 'sand', 'Steelbars', 'Paint')
materials[0]: cementbags
materials[1:3]: ('bricks', 'sand')
```